ECON2105

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Chapter 1

Sampling

Definition 1.0.1: parameter

The parameter(s) of a distribution of a random variable is a constant, the knowledge of which completely determines the distribution.

Example.

- Example 1: $X \sim \mathit{Normal}(\mu, \sigma^2)$ has 2 parameters
- Example 2: $X \sim \mathit{Binomial}(n, p)$ has 2 parameters
- Example 3: $X \sim t_d$ has 1 parameter

Usually we do not know the true values of the parameters and our goal is to try and estimate them from data

Definition 1.0.2: random sample

A random sample of size n is n i.i.d. random variables $X_1, X_2, \ldots, X_n$.

Since $X_1, X_2, \ldots, X_n$ are i.i.d., they all have the same distribution, hence the **same parameters**. A random sample can hence be used to **estimate the parameters**.

Due to **independence**, the joint density/probability function of $X_1, X_2, \ldots, X_n$ is the product of their marginal density/probability functions. Independence can be relaxed to deal with dependent data (e.g. time-series) (beyond the scope of this course).

Identical distribution can also be relaxed. We will sometimes do this. *Data* are a particular realisation of the random sample, denoted $x_1, x_2, \ldots, x_n$.

Definition 1.0.3: statistic

A *statistic* is a real valued function of a random sample.

Example.

- Example 1: $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ (sample mean)
- Example 2: $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$ (sample variance)
- Example 3: $\tilde{s}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - E(X))^2$ (sample variance with known mean)
- Example 4: $\min(X_1, X_2, \ldots, X_n), \max(X_1, X_2, \ldots, X_n)$ (sample minimum and maximum)
- Example 5: $\frac{1}{n}\sum_{i=1}^{n} X_i^m$, $\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^m$ (m^{th} sample moment)

Given data, we can compute a realisation of any statistic by replacing $X_1, X_2, \ldots, X_n$ with $x_1, x_2, \ldots, x_n$.

Definition 1.0.4: sampling distribution

Statistics are a function of n random variables, hence any statistic is itself a random variable. Since a statistic is a random variable, it has a distribution. Its distribution will typically depend on the sample size n.
We call this distribution the *sampling distribution* or *small sample distribution* of the statistic.

Definition 1.0.5: asymptotic distribution

As the sample gets large ($n \to \infty$), the sampling distribution often converges to a known distribution (typically a normal distribution).
We call this distribution the *asymptotic distribution* or *large sample distribution* of the statistic.

1.1 Sampling distribution of $\overline{X}$ and s

Let $X_1, X_2, \ldots, X_n$ be a random sample with $E(X_i) = \mu$ and $V(X_i) = \sigma^2$ for $i = 1, \ldots, n$.

Theorem 1.1.1

the sampling distribution of $\overline{X}$ verifies

$$E(\overline{X}) = \mu, \quad V(\overline{X}) = \frac{\sigma^2}{n}$$

- *Notice that $V(\overline{X})$ depends on n.*
- *The $X_1, \ldots, X_n$ need not be identically distributed, but they must be independent and have the same expectation (μ) and variance (σ^2).*

Proof. Modify the proof for the case in which $E(X_i) = \mu_i$ and $V(X_i) = \sigma_i^2$ for $i = 1, \ldots, n$.
Starting with the sample mean,

$$E(\overline{X}) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}E\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n}\sum_{i=1}^{n} \mu_i = \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{1}{n}n\mu = \mu$$

Now for the variance,

$$V(\overline{X}) = V\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}V\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\left(\sum_{i=1}^{n} V(X_i) + \sum_{i=1}^{n}\sum_{\substack{j=1 \\ j\neq i}}^{n} \mathrm{Cov}(X_i, X_j)\right)$$

Since the X_i are independent, $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$, so

$$= \frac{1}{n^2}\sum_{i=1}^{n} V(X_i) = \frac{1}{n^2}\sum_{i=1}^{n} \sigma_i^2$$

If $\sigma_i^2 = \sigma^2$ for all i, then

$$= \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 = \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n}$$

□

Theorem 1.1.2

the sampling distribution of s^2 verifies $E(s^2) = \sigma^2$.

Proof. Now for the sample variance:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i^2 - 2X_i\overline{X} + \overline{X}^2)$$

$$= \frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2 - 2\sum_{i=1}^{n}X_i\overline{X} + \sum_{i=1}^{n}\overline{X}^2\right) = \frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2 - 2\overline{X}\sum_{i=1}^{n}X_i + n\overline{X}^2\right)$$

$$= \frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2 - 2n\overline{X}^2 + n\overline{X}^2\right) = \frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2 - n\overline{X}^2\right)$$

Recall

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n}X_i^2 - n\overline{X}^2\right)$$

Hence,

$$E(s^2) = \frac{1}{n-1}E\left(\sum_{i=1}^{n}X_i^2 - n\overline{X}^2\right) = \frac{1}{n-1}\left(\sum_{i=1}^{n}E(X_i^2) - nE(\overline{X}^2)\right)$$

Now use

$$E(X_i^2) = V(X_i) + (E(X_i))^2 = \sigma^2 + \mu^2$$

$$E(\overline{X}^2) = V(\overline{X}) + (E(\overline{X}))^2 = \frac{\sigma^2}{n} + \mu^2$$

Hence,

$$E(s^2) = \frac{1}{n-1}\left(\sum_{i=1}^{n}(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right)$$

$$= \frac{1}{n-1}\left(n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right)$$

$$= \frac{n}{n-1}\left(\sigma^2 - \frac{\sigma^2}{n}\right)$$

$$= \frac{n}{n-1} \times \frac{n-1}{n}\sigma^2 = \sigma^2$$

□

1.2 Sampling from the normal distribution

We already showed that if $X_i \sim Normal(\mu, \sigma^2)$ for $i = 1, \ldots, n$ are i.i.d.,

$$\overline{X} \sim Normal\left(\mu, \frac{\sigma^2}{n}\right)$$

So if $Z_i \sim Normal(0, 1)$ for $i = 1, \ldots, n$ are i.i.d., then

$$\overline{Z} \sim Normal\left(0, \frac{1}{n}\right)$$

Lemma 1.2.1

$\overline{Z}$ and $\sum_{i=1}^{n}(Z_i - \overline{Z})^2$ are independent and that

$$\sum_{i=1}^{n}(Z_i - \overline{Z})^2 \sim \chi^2_{n-1}$$

Proof. Since $Z_i \sim Normal(0,1)$ and $\overline{Z} \sim Normal(0, 1/n)$,

$$Z_i - \overline{Z} = \frac{n-1}{n}Z_i - \frac{1}{n}\sum_{\substack{j=1\\ j\neq i}}^{n} Z_j \sim Normal\left(0, \frac{n-1}{n}\right)$$

We also have that for $i = 1, \ldots, n$,

$$\mathrm{Cov}(Z_i, \overline{Z}) = E(Z_i\overline{Z}) - E(Z_i)E(\overline{Z}) = E(Z_i\overline{Z}) = \frac{1}{n}E(Z_i^2) = \frac{1}{n}$$

and

$$\begin{aligned}\mathrm{Cov}(Z_i - \overline{Z}, \overline{Z}) &= E\left((Z_i - \overline{Z})\overline{Z}\right) - E(Z_i - \overline{Z})E(\overline{Z}) = E(Z_i\overline{Z}) - E(\overline{Z}^2)\\ &= \frac{1}{n} - \mathrm{Var}(\overline{Z}) = \frac{1}{n} - \frac{1}{n} = 0\end{aligned}$$

By similar arguments (check yourself!) we have

$$\mathrm{Cov}(Z_i - \overline{Z}, Z_j - \overline{Z}) = -\frac{1}{n}$$

Now we need to use the *multivariate normal distribution* (just for this proof) and to keep things simple, let $n = 3$. Since $\overline{Z}$ and Z_i are normally distributed, the random vector

$$\begin{pmatrix} \overline{Z} \\ Z_1 - \overline{Z} \\ Z_2 - \overline{Z} \\ Z_3 - \overline{Z} \end{pmatrix} \sim Normal\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{n} & 0 & 0 & 0 \\ 0 & \frac{n-1}{n} & -\frac{1}{n} & -\frac{1}{n} \\ 0 & -\frac{1}{n} & \frac{n-1}{n} & -\frac{1}{n} \\ 0 & -\frac{1}{n} & -\frac{1}{n} & \frac{n-1}{n} \end{pmatrix}\right)$$

Recall

$$\begin{pmatrix} \overline{Z} \\ Z_1 - \overline{Z} \\ Z_2 - \overline{Z} \\ Z_3 - \overline{Z} \end{pmatrix} \sim Normal\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{n} & 0 & 0 & 0 \\ 0 & \frac{n-1}{n} & -\frac{1}{n} & -\frac{1}{n} \\ 0 & -\frac{1}{n} & \frac{n-1}{n} & -\frac{1}{n} \\ 0 & -\frac{1}{n} & -\frac{1}{n} & \frac{n-1}{n} \end{pmatrix}\right)$$

Due to the zeros in the covariance matrix, this means that $\overline{Z}$ is independent of

$$\begin{pmatrix} Z_1 - \overline{Z} \\ Z_2 - \overline{Z} \\ Z_3 - \overline{Z} \end{pmatrix}$$

Hence, $\overline{Z}$ is independent of $\sum_{i=1}^{n}(Z_i - \overline{Z})^2$.

Let us now show that $\sum_{i=1}^{n}(Z_i - \overline{Z})^2 \sim \chi^2_{n-1}$. We start with

$$\begin{aligned}W = \sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n}(Z_i - \overline{Z} + \overline{Z})^2 &= \sum_{i=1}^{n}(Z_i - \overline{Z})^2 + \sum_{i=1}^{n}\overline{Z}^2 + 2\overline{Z}\sum_{i=1}^{n}(Z_i - \overline{Z})\\ &= \sum_{i=1}^{n}(Z_i - \overline{Z})^2 + n\overline{Z}^2\end{aligned}$$

and we know that $W \sim \chi^2_n$.

Also, we know that $\overline{Z} \sim Normal(0, 1/n)$, hence $\sqrt{n}\overline{Z} \sim Normal(0, 1)$, hence

$$n\overline{Z}^2 \sim \chi_1^2$$

Recall

$$W = \sum_{i=1}^{n}(Z_i - \overline{Z})^2 + n\overline{Z}^2 \sim \chi_n^2, \quad n\overline{Z}^2 \sim \chi_1^2$$

Now let us compute the moment generating function:

$$m_W(t) = E\left(\exp\left(t\sum_{i=1}^{n}(Z_i - \overline{Z})^2 + tn\overline{Z}^2\right)\right)$$

$$= E\left(\exp\left(t\sum_{i=1}^{n}(Z_i - \overline{Z})^2\right)\exp\left(tn\overline{Z}^2\right)\right)$$

$$= E\left(\exp\left(t\sum_{i=1}^{n}(Z_i - \overline{Z})^2\right)\right)E\left(\exp\left(tn\overline{Z}^2\right)\right) \quad \text{(independence)}$$

$$= m_{\sum_{i=1}^{n}(Z_i-\overline{Z})^2}(t) \times m_{n\overline{Z}^2}(t)$$

Recall again

$$W = \sum_{i=1}^{n}(Z_i - \overline{Z})^2 + n\overline{Z}^2 \sim \chi_n^2, \quad n\overline{Z}^2 \sim \chi_1^2$$

So we know that

$$m_W(t) = (1-2t)^{-n/2}, \quad m_{n\overline{Z}^2}(t) = (1-2t)^{-1/2}, \quad t < 1/2$$

Hence,

$$(1-2t)^{-n/2} = m_{\sum_{i=1}^{n}(Z_i-\overline{Z})^2}(t)(1-2t)^{-1/2}$$

Rearranging gives

$$m_{\sum_{i=1}^{n}(Z_i-\overline{Z})^2}(t) = (1-2t)^{-(n-1)/2}, \quad t < 1/2$$

which is the moment generating function of χ_{n-1}^2

□

Theorem 1.2.2

If $X_i \sim Normal(\mu, \sigma^2)$ are independent, then the previous lemma implies that $\overline{X}$ is independent of s^2 and that

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

Chapter 2

Inference

Inference refers to estimation, confidence intervals and hypothesis testing on parameters using a random sample.

In general, we refer to one or more parameters denoted by θ. The parameter θ lies in a parameter space Θ (i.e. $\theta \in \Theta$). The parameter space encodes any restrictions on θ.

Example.

> For example, for $Normal(\mu, \sigma^2)$,
>
> $$\theta = (\mu, \sigma^2) \in (-\infty, \infty) \times (0, \infty)$$

Definition 2.0.1

An *estimator* of θ is a statistic used to estimate θ. We use $\hat{\theta}$ to denote an estimator of θ.

Example.

> For example, $\overline{X}$ is an estimator of $E(X) = \mu$.

$\hat{\theta}$ is therefore a function of the random sample, hence it is a random variable.

Definition 2.0.2

An *estimate* of θ is a particular realization of $\hat{\theta}$.

Given data (i.e. a realisation of a random sample), we can compute an estimate.

Example.

> For example, the average of the data,
>
> $$\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i,$$
>
> is a realisation of $\overline{X}$.

Finite sample properties concern the **sampling distribution** of $\hat{\theta}$.

- *Unbiasedness*: $E(\hat{\theta}) = \theta$?
- *Efficiency*: $V(\hat{\theta})$ is 'small'?
- *Mean squared error*:
  $$E((\hat{\theta} - \theta)^2) = (E(\hat{\theta}) - \theta)^2 + V(\hat{\theta})$$
  is 'small'?

- *Known distribution*: The sampling distribution of $\hat{\theta}$ is known (e.g. Normal)?

Example.

> Show that
> $$E((\hat{\theta}-\theta)^2) = (E(\hat{\theta})-\theta)^2 + V(\hat{\theta}).$$

Example.

> Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from some distribution with expectation μ and variance σ^2. Compute the MSE of
> $$\hat{\mu}_1 = \bar{X} \quad \text{and} \quad \hat{\mu}_2 = \frac{n-1}{n}\bar{X}.$$

We already showed that $\mathbb{E}(\hat{\mu}_1) = \mathbb{E}(\bar{X}) = \mu$, hence

$$\text{MSE}(\hat{\mu}_1) = \mathbb{V}(\hat{\mu}_1) = \mathbb{V}(\bar{X}) = \frac{\sigma^2}{n}.$$

Since $\hat{\mu}_2 = \frac{n-1}{n}\hat{\mu}_1$, we have

$$\mathbb{E}(\hat{\mu}_2) = \frac{n-1}{n}\mu, \quad \mathbb{V}(\hat{\mu}_2) = \left(\frac{n-1}{n}\right)^2 \frac{\sigma^2}{n}.$$

Hence,

$$\text{MSE}(\hat{\mu}_2) = \left(\frac{n-1}{n}\mu - \mu\right)^2 + \left(\frac{n-1}{n}\right)^2 \frac{\sigma^2}{n} = \frac{\mu^2}{n^2} + \left(\frac{n-1}{n}\right)^2 \frac{\sigma^2}{n}.$$

Thus, $\hat{\mu}_1$ is unbiased but has larger variance, and $\hat{\mu}_2$ is biased but has smaller variance.

Which has the smaller MSE depends on μ, σ^2, and n. For example, if $\mu = 0$ then

$$\text{MSE}(\hat{\mu}_2) < \text{MSE}(\hat{\mu}_1).$$

Asymptotic properties concern the **asymptotic distribution** of $\hat{\theta}$.

- *Asymptotic Unbiasedness*: $\lim_{n\to\infty} E(\hat{\theta}) = \theta$?
- *Asymptotic Efficiency*: $\lim_{n\to\infty} V(\hat{\theta})$ is 'small'?
- *Mean squared error*:
  $$\lim_{n\to\infty} E\left((\hat{\theta}-\theta)^2\right) = \lim_{n\to\infty}\left((E(\hat{\theta})-\theta)^2 + V(\hat{\theta})\right)$$
  is 'small'?
- *Consistency*: $\hat{\theta} \to \theta$ as $n \to \infty$?
- *Known distribution*: The asymptotic distribution of $\hat{\theta}$ is known? This is typically true, and either normal or χ^2.

Example.

> Are the estimators asymptotically unbiased? What is the asymptotic MSE of the estimators?

Recall

$$\mathbb{E}(\hat{\mu}_1) = \mu, \quad \mathbb{E}(\hat{\mu}_2) = \frac{n-1}{n}\mu,$$

$$\mathbb{V}(\hat{\mu}_1) = \frac{\sigma^2}{n}, \quad \mathbb{V}(\hat{\mu}_2) = \left(\frac{n-1}{n}\right)^2 \frac{\sigma^2}{n},$$

and

$$\text{MSE}(\hat{\mu}_1) = \frac{\sigma^2}{n},$$

$$\text{MSE}(\hat{\mu}_2) = \frac{\mu^2}{n^2} + \left(\frac{n-1}{n}\right)^2 \frac{\sigma^2}{n}.$$

So both estimators are asymptotically unbiased and both have asymptotic MSE equal to zero.

Definition 2.0.3: Estimation strategies

- *Method of moments*: To estimate q parameters, match q sample moments with q moments.
- *General method of moments*: Match $p > q$ sample moments with p moments.
- *Maximum likelihood*: Find the parameters which maximize the probability of the data.
- *Least squares*: 'Fit' the data as closely as possible by minimizing the sum of squared errors.

2.1 Method of moments

Method of moments uses that the moments are functions of parameters. For simplicity we focus on moments around the origin. The method of moments uses as many moments as parameters to be estimated.

Example.

For $X \sim \mathit{Normal}(\mu, \sigma^2)$,

$$E(X) = \mu, \quad E(X^2) = \mu^2 + \sigma^2$$

Example.

For $X \sim \mathit{Poisson}(\lambda)$,

$$E(X) = \lambda$$

We set moments equal to sample moments

Example.

For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\frac{1}{n}\sum_{i=1}^{n} X_i = \hat{\mu}, \quad \frac{1}{n}\sum_{i=1}^{n} X_i^2 = \hat{\mu}^2 + \hat{\sigma}^2$$

Hence,

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2$$

Example.

For $X \sim \text{Poisson}(\lambda)$,

$$\frac{1}{n}\sum_{i=1}^{n} X_i = \hat{\lambda}$$

If we use more moments than parameters we call it the General method of moments (beyond the scope of this course).

The method of moments works because the sample moments 'converge' to the moments as $n \to \infty$ (more on this later).

Example.

As $n \to \infty$, for $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \to \quad \mathbb{E}(X) = \mu$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2 \quad \to \quad \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \sigma^2$$

Example.

Find the method of moments estimator for the Gamma(α, β) parameters.

We know that $\mathbb{E}(X) = \alpha\beta$ and $\mathbb{E}(X^2) = \alpha\beta^2 + \alpha^2\beta^2$, hence the method of moments estimators are

$$\hat{\alpha}\hat{\beta} = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad \hat{\alpha}\hat{\beta}^2 + \hat{\alpha}^2\hat{\beta}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$$

Now let's define $\hat{\mu} = \hat{\alpha}\hat{\beta}$ and $\hat{\sigma}^2 = \hat{\alpha}\hat{\beta}^2$ so that

$$\hat{\sigma}^2 = \left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right) - \hat{\mu}^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

where we showed the final equality in the second proof of the Sampling lecture.

Now use

$$\hat{\beta} = \frac{\hat{\alpha}\hat{\beta}^2}{\hat{\alpha}\hat{\beta}} = \frac{\hat{\sigma}^2}{\hat{\mu}}$$

$$\hat{\alpha} = \frac{(\hat{\alpha}\hat{\beta})^2}{\hat{\alpha}\hat{\beta}^2} = \frac{\hat{\mu}^2}{\hat{\sigma}^2}$$

And recall

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Then,

$$\hat{\alpha}_{\text{MoM}} = \frac{\bar{X}^2}{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2}, \quad \hat{\beta}_{\text{MoM}} = \frac{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2}{\bar{X}}$$

2.2 Asymptotic theory

We typically know little about the sampling (small sample) distribution of estimators. However, as we will see, we know a lot about the large sample (asymptotic as $n \to \infty$) distribution of estimators. This is not directly useful, because $n \to \infty$ cannot be satisfied in real data. However, we can still use it to approximate the small sample distribution, allowing us to perform hypothesis tests, construct confidence intervals etc. The larger is n, the better the approximation. We now study several notions of *convergence*.

2.2.1 Converges in probability

Definition 2.2.1

A sequence of real numbers $a_1, a_2, \ldots$ converges to c if

$$\lim_{n\to\infty} a_n = c$$

This means that for any $\delta > 0$ there exists N such that

$$|a_n - c| < \delta, \quad \forall n \geq N$$

We are interested instead in a sequence of random variables $X_1, X_2, \ldots$

Definition 2.2.2

A sequence of random variables $X_1, X_2, \ldots$ *converges in probability* to c if for any $\epsilon > 0$

$$\lim_{n\to\infty} \text{Prob}(|X_n - c| > \epsilon) = 0$$

- *We use the notation $X_n \xrightarrow{p} c$ or equivalently plim $X_n = c$*
- *Convergence in probability is difficult to check in practice*

Definition 2.2.3

An estimator $\hat{\theta}$ is *consistent* if $\hat{\theta} \xrightarrow{p} \theta$.

Consistency is a basic requirement for any estimator

Definition 2.2.4

A sequence of random variables $X_1, X_2, \ldots$ *converges in quadratic mean* to c if

$$\lim_{n\to\infty} E(X_n) = c, \qquad \lim_{n\to\infty} V(X_n) = 0$$

- *We use the notation $X_n \xrightarrow{qm} c$*
- *Convergence in quadratic mean is straightforward to check*

Theorem 2.2.5

If $X_n \xrightarrow{qm} c$ then $X_n \xrightarrow{p} c$

Example.

Show consistency of the sample mean of a random sample.

We have previously shown

$$E(\overline{X}) = \mu, \qquad V(\overline{X}) = \frac{\sigma^2}{n}$$

So

$$\lim_{n\to\infty} E(\overline{X}) = \mu, \qquad \lim_{n\to\infty} V(\overline{X}) = 0$$

Hence

$$\overline{X} \xrightarrow{qm} \mu \quad \text{which implies} \quad \overline{X} \xrightarrow{p} \mu$$

Theorem 2.2.6

For any continuous function $g(\cdot)$ which is not a function of n,

$$\text{plim } g(X_n) = g(\text{plim } X_n)$$

Here p means convergence in probability.

Theorem 2.2.7

If $X_n \xrightarrow{p} c$ and $Y_n \xrightarrow{p} d$ then

- $X_n + Y_n \xrightarrow{p} c + d$
- $X_n Y_n \xrightarrow{p} cd$
- $X_n / Y_n \xrightarrow{p} c/d$ if $d \neq 0$

Theorem 2.2.8: Weak law of large numbers

Let $X_1, X_2, \ldots, X_n$ be random variables verifying, for $i = 1, \ldots, n$,

$$-\infty < \mu_i < \infty, \quad 0 < \sigma_i^2 < \infty, \quad \text{Cov}(X_i, X_j) = 0 \text{ for } j \neq i$$

and

$$\lim_{n \to \infty} \frac{1}{n} \sum_{i=1}^{n} \sigma_i^2 < \infty$$

Then

$$\text{plim} \left(\overline{X} - \frac{1}{n} \sum_{i=1}^{n} \mu_i \right) = 0$$

- *WLLN is useful because it concerns convergence of averages to expectations*
- *Notice that we don't require independence nor identically distributed, so WLLN is very general*
- *If $X_1, \ldots, X_n$ are i.i.d. (e.g. a random sample), $-\infty < \mu < \infty$ and $0 < \sigma^2 < \infty$ then $\overline{X} \xrightarrow{p} \mu$*

Proposition 2.2.9

For i.i.d. random variables $X_1, X_2, \ldots, X_n$ and $Y_i = g(X_i)$ for any function $g(\cdot)$, if $E(Y)$ and $\text{Var}(Y)$ are finite then

$$\frac{1}{n} \sum_{i=1}^{n} Y_i \xrightarrow{p} E(Y)$$

Example.

For example, $g(X) = X^m$, then $\frac{1}{n} \sum_{i=1}^{n} X_i^m \xrightarrow{p} E(X^m)$.

Example.

Given a random sample, show consistency of the MLE of σ^2 for $X \sim \text{Normal}(\mu, \sigma^2)$.

Proof. From the tutorial,

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}^2$$

By WLLN,

$$\overline{X} \xrightarrow{p} E(X) = \mu, \qquad \frac{1}{n}\sum_{i=1}^{n} X_i^2 \xrightarrow{p} E(X^2) = \mu^2 + \sigma^2$$

Now let $g(\overline{X}) = \overline{X}^2$. Since $g(\cdot)$ is continuous and does not depend on n,

$$\text{plim } \overline{X}^2 = (\text{plim } \overline{X})^2 = \mu^2$$

So,

$$\text{plim } \hat{\sigma}^2 = \text{plim}\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right) - \text{plim}(\overline{X}^2) = \mu^2 + \sigma^2 - \mu^2 = \sigma^2$$

□

2.2.2 Converges in distribution

Notions of convergence to a constant are useful for establishing the limit of an estimator, hence consistency. However, they don't tell us anything about its asymptotic distribution, which we need for hypothesis testing and confidence intervals. For this reason we also consider convergence to a random variable.

Definition 2.2.10

A sequence of random variables $X_1, X_2, \ldots$ *converges in m^{th} mean* to a random variable X if

$$\lim_{n\to\infty} E(|X_n - X|^m) = 0$$

If $m = 2$ this is convergence in quadratic mean

Theorem 2.2.11

If $E(|X|^m)$ is finite then $\lim_{n\to\infty} E(|X_n|^m) = E(|X|^m)$

Definition 2.2.12

A sequence of random variables $X_1, X_2, \ldots$ *converges in distribution* to a random variable X if for all x being continuous points of $F_X(\cdot)$,

$$\lim_{n\to\infty} |F_{X_n}(x) - F_X(x)| = 0$$

We use the notation $X_n \xrightarrow{d} X$.

- *If the distribution of X is known (e.g., Normal$(0,1)$) we use the shorthand*

$$X_n \xrightarrow{d} \textit{Normal}(0,1)$$

  which means $X_n \xrightarrow{d} X$ and $X \sim$ Normal$(0,1)$

- *To show convergence in distribution it suffices to show*

$$\lim_{n\to\infty} m_{X_n}(t) = m_X(t), \quad \forall t$$

Theorem 2.2.13

If $X_n \xrightarrow{d} X$ then for any continuous function $g(\cdot)$,

$$g(X_n) \xrightarrow{d} g(X)$$

Theorem 2.2.14

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} d$ then

- $X_n + Y_n \xrightarrow{d} X + d$
- $X_n Y_n \xrightarrow{d} Xd$
- $X_n / Y_n \xrightarrow{d} X/d$ if $d \neq 0$

Theorem 2.2.15: Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with

$$-\infty < \mu < \infty, \quad 0 < \sigma^2 < \infty.$$

Then

$$\sqrt{n}(\overline{X} - \mu) \xrightarrow{d} \text{Normal}(0, \sigma^2)$$

If $X_i \sim$ Normal(μ, σ^2) for $i = 1, \ldots, n$ then we already knew $\sqrt{n}(\overline{X} - \mu) \sim$ Normal$(0, \sigma^2)$ because $\overline{X} \sim$ Normal$(\mu, \sigma^2/n)$. This is true whether n is small or large. CLT tells us that this is true for ***any*** *distribution as $n \to \infty$. So we can always use the normal distribution for hypothesis testing and confidence intervals if n is sufficiently large.*

Definition 2.2.16

If $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \text{Normal}(0, \sigma^2_{\hat{\theta}})$ then the *asymptotic distribution* is

$$\hat{\theta} \overset{a}{\sim} \text{Normal}\left(\theta, \frac{\sigma^2_{\hat{\theta}}}{n}\right)$$

We can use this to approximate the sampling distribution of $\hat{\theta}$ for the purposes of hypothesis testing and confidence intervals. The larger is n, the better is the approximation.

Given a random sample $X_1, \ldots, X_n$ from **any** distribution, we have the approximation

$$\overline{X} \overset{a}{\sim} \text{Normal}\left(\mu, \frac{\sigma^2}{n}\right)$$

If the sample is from a normal distribution, the approximation is exact because

$$\overline{X} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{n}\right)$$

So when we use '$\overset{a}{\sim}$', it means that we are using an asymptotic approximation, and when we use '$\sim$' the approximation is exact.

Theorem 2.2.17: Delta method

Suppose that

$$\sqrt{n}(X_n - \mu) \xrightarrow{d} \text{Normal}(0, \sigma^2)$$

and $g(\cdot)$ is a continuously differentiable function with $g'(\mu) \neq 0$ then

$$\sqrt{n}(g(X_n) - g(\mu)) \xrightarrow{d} \text{Normal}(0, \sigma^2[g'(\mu)]^2)$$

The Delta method allows us to construct the asymptotic distribution of a function of a random variable.

Example.

> So if $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \text{Normal}(0, \sigma^2_{\hat{\theta}})$

then

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} \text{Normal}(0, [g'(\theta)]^2\sigma^2_{\hat{\theta}})$$

and its asymptotic distribution is

$$g(\hat{\theta}) \overset{a}{\sim} \text{Normal}\left(g(\theta), \frac{[g'(\theta)]^2\sigma^2_{\hat{\theta}}}{n}\right)$$

Example.

> Find the asymptotic distribution of $\hat{\sigma}$

From the tutorial,

$$I(\theta) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix} \quad \text{and hence} \quad I(\theta)^{-1} = \begin{pmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{pmatrix}$$

So by the ML properties,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \text{Normal}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{bmatrix}\right)$$

Hence,

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} \text{Normal}(0, 2\sigma^4)$$

and let $g(\hat{\sigma}^2) = \hat{\sigma}$ (i.e. $g(x) = \sqrt{x}$ for $x > 0$).

Since $g(\cdot)$ is differentiable with $g'(x) = \frac{1}{2\sqrt{x}}$ and $g'(\sigma) \neq 0$, then by the Delta method

$$\sqrt{n}(\hat{\sigma} - \sigma) \xrightarrow{d} \text{Normal}\left(0, \frac{2\sigma^4}{4\sigma^2}\right) = \text{Normal}\left(0, \frac{\sigma^2}{2}\right)$$

So the asymptotic distribution is

$$\hat{\sigma} \overset{a}{\sim} \text{Normal}\left(\sigma, \frac{\sigma^2}{2n}\right)$$

2.3 Ordinary least squares

OLS is an estimator for the parameters of a *regression model*.

Definition 2.3.1

A regression model is a model for $E(Y \mid X)$, where Y represents an outcome of interest (e.g. wages) and X is a regressor (e.g. education).

We observe the bi-variate random sample

$$(Y_1, X_1), (Y_2, X_2), \ldots, (Y_n, X_n)$$

The basic linear regression model comprises an outcome equation

$$Y_i = \beta_1 + \beta_2 X_i + U_i, \quad i = 1, \ldots, n$$

and assumptions on the *error* U_i.

We will consider the assumptions for $i = 1, \ldots, n$:

$$E(U_i \mid X_i) = 0 \quad \text{(exogeneity)}$$

$$E(U_i^2 \mid X_i) = \sigma^2 \quad \text{(homoskedasticity)}$$

$$E(U_i U_j \mid X_i, X_j) = 0, \quad j \neq i \quad \text{(no serial correlation)}$$

From the assumptions above, by the law of iterated expectations we obtain

Proposition 2.3.2

Error properties:

$$\begin{aligned} E(U_i) &= 0 \\ E(U_i X_i) &= 0 \\ E(U_i^2) &= \sigma^2 \\ E(U_i U_j) &= 0, \quad j \neq i \end{aligned}$$

for $i = 1, \ldots, n$.

Theorem 2.3.3

Using the outcome equation and the exogeneity assumption, we immediately obtain

$$\mathbb{E}(Y_i \mid X_i) = \beta_1 + \beta_2 X_i$$

for $i = 1, \ldots, n$.

So a regression models the conditional expectation of the outcome as a linear function of the regressors with weights given by the parameters β_1, β_2.

Since

$$\mathbb{E}(Y_i \mid X_i) = \beta_1 + \beta_2 X_i$$

we can interpret β_2 as the *marginal effect* of X_i on $\mathbb{E}(Y_i \mid X_i)$. That is,

$$\beta_2 = \frac{\partial \mathbb{E}(Y_i \mid X_i)}{\partial X_i}$$

Equivalently,

$$\beta_2 = \mathbb{E}(Y_i \mid X_i + 1) - \mathbb{E}(Y_i \mid X_i)$$

So β_2 is the change in the expected outcome (e.g., wages) when the regressor (e.g., years of education) increases by one unit. However, β_1 does not have a meaningful interpretation in general.

2.3.1 Computing OLS

The OLS estimator is computed by minimizing the *sum of squared errors*. Formally, $\hat{\beta}_1, \hat{\beta}_2$ are obtained by minimizing.

$$\sum_{i=1}^{n} U_i^2$$

with respect to β_1 and β_2.

Example.

Derive the OLS estimators of β_1 and β_2.

Proof. Objective:

$$\min_{\beta_1,\beta_2} \sum_{i=1}^{n} (y_i - \mathbb{E}(y_i \mid X_i))^2$$

which implies

$$\min_{\beta_1,\beta_2} \sum_{i=1}^{n} (y_i - \beta_1 - \beta_2 X_i)^2$$

First order conditions:

$$\frac{\partial}{\partial \beta_1} = -2 \sum_{i=1}^{n} \left(y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i \right) = 0 \tag{1}$$

$$\frac{\partial}{\partial \beta_2} = -2 \sum_{i=1}^{n} X_i \left(y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i \right) = 0 \tag{2}$$

From (1):

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$$

From (1):

$$\bar{x} \sum_{i=1}^{n} (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) = 0 \tag{3}$$

From (2):

$$\sum_{i=1}^{n} x_i (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) = 0 \tag{4}$$

Now (4) minus (3) gives:

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) = 0$$

Substituting $\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$ into the equation:

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y} + \hat{\beta}_2 \bar{x} - \hat{\beta}_2 x_i) = 0$$

Rearranging:

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x}) \hat{\beta}_2 (x_i - \bar{x})$$

Thus:

$$\hat{\beta}_2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Now, using

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$$

we can write:

$$\hat{\beta}_1 = \bar{y} - \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \bar{x}$$

□

Example.

Compute the OLS estimator of β_2 when $\beta_1 = 0$.

We have $Y_i = \beta X_i + U_i$ and we need to minimize

$$\sum_{i=1}^{n}(Y_i - \beta X_i)^2$$

Differentiating yields

$$\frac{\partial}{\partial \beta}\sum_{i=1}^{n}(Y_i - \beta X_i)^2 = -2\sum_{i=1}^{n} X_i(Y_i - \beta X_i)$$

so $\hat{\beta}$ satisfies

$$-2\sum_{i=1}^{n} X_i(Y_i - \hat{\beta} X_i) = 0$$

Expanding,

$$\sum_{i=1}^{n}(Y_i X_i - \hat{\beta} X_i^2) = 0$$

$$\sum_{i=1}^{n} Y_i X_i - \hat{\beta}\sum_{i=1}^{n} X_i^2 = 0$$

thus

$$\hat{\beta} = \frac{\sum_{i=1}^{n} Y_i X_i}{\sum_{i=1}^{n} X_i^2}$$

Theorem 2.3.4

The OLS estimator $\hat{\beta}_1, \hat{\beta}_2$ is also the MM estimator with moments

$$\mathbb{E}(U_i) = 0, \quad \mathbb{E}(X_i U_i) = 0$$

which we derived earlier.

Proof. The OLS first order conditions are

$$\sum_{i=1}^{n}(Y_i - \hat{\beta}_1 - X_i\hat{\beta}_2) = 0, \quad \sum_{i=1}^{n} X_i(Y_i - \hat{\beta}_1 - X_i\hat{\beta}_2) = 0$$

Multiplying by $\frac{1}{n}$ yields

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{\beta}_1 - X_i\hat{\beta}_2) = 0, \quad \frac{1}{n}\sum_{i=1}^{n} X_i(Y_i - \hat{\beta}_1 - X_i\hat{\beta}_2) = 0$$

□

We now consider a particular case in which it is known that $\beta_2 = 0$. Then we know that $\mathbb{E}(Y_i \mid X_i) = \beta_1$, so by iterated expectations $\mathbb{E}(Y_i) = \beta_1$. So in this case β_1 is the expectation of Y_i for $i = 1, \ldots, n$...so let us call it μ instead, hence

$$Y_i = \mu + U_i$$

Now, the OLS estimator of μ is obtained by minimizing

$$\sum_{i=1}^{n} U_i^2 = \sum_{i=1}^{n}(Y_i - \mu)^2$$

with respect to μ.

We do this by differentiating with respect to μ and setting it equal to zero. Differentiating yields

$$\frac{\partial \sum_{i=1}^{n}(Y_i - \mu)^2}{\partial \mu} = -2\sum_{i=1}^{n}(Y_i - \mu)$$

Then the estimator $\hat{\mu}$ satisfies

$$-2\sum_{i=1}^{n}(Y_i - \hat{\mu}) = 0$$

$$\sum_{i=1}^{n} Y_i - n\hat{\mu} = 0$$

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} Y_i = \overline{Y}$$

2.3.2 OLS is BLUE

An important result is that the OLS estimator is the *Best Linear Unbiased Estimator*. This means that OLS is the Linear Unbiased estimator with the smallest possible variance.

Recall. *Unbiasedness*: $E(\hat{\theta}) = \theta$

Proof. omitted. □

Theorem 2.3.5

OLS is

- BLUE: $E(\hat{\beta}_1) = \beta_1, E(\hat{\beta}_2) = \beta_2$
- Consistent
- Asymptotically normally distributed

2.4 Maximum Likelihood

Suppose that we have a random sample $X_1, X_2, \ldots, X_n$ and data $x_1, x_2, \ldots, x_n$. Then the joint density of the random sample, evaluated at the data $x_1, \ldots, x_n$ is

$$f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i) \quad \text{(independent)}$$

$$= \prod_{i=1}^{n} f_X(x_i) \quad \text{(identically distributed)}$$

$$= L(\theta \mid x_1, \ldots, x_n)$$

which depends on the parameter(s) θ through $f_X(\cdot)$.

Definition 2.4.1

The maximum likelihood **estimate** finds the maximum value of $L(\theta \mid x_1, \ldots, x_n)$. It maximizes the likelihood of the data we observe.

Definition 2.4.2

The maximum likelihood **estimator** maximizes the *likelihood function*

$$L(\theta \mid X_1, \ldots, X_n) = \prod_{i=1}^{n} f_X(X_i)$$

with respect to θ.

This is challenging due to the product, which is difficult to differentiate. So instead we maximize the *log likelihood function*

$$\ln L(\theta \mid X_1, \ldots, X_n) = \sum_{i=1}^{n} \ln[f_X(X_i)]$$

which yields the same $\hat{\theta}$ because $\ln(\cdot)$ is strictly increasing.

2.4.1 Computing ML

We can usually do this analytically, but for some $f_X(\cdot)$ we have to resort to numerical methods.

- If $f_X(\cdot)$ is differentiable, then the maximum likelihood estimator solves the *score* equations:

$$\left.\frac{\partial \ln L}{\partial \theta}\right|_{\hat{\theta}} = 0$$

- Otherwise, we use numerical methods such as the Newton-Raphson algorithm.
  If there is just one parameter θ, set a starting value θ_0 and then iterate

$$\theta_{s+1} = \theta_s - \left(\frac{\partial \ln L}{\partial \theta}\right)\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)^{-1}\Bigg|_{\theta_s}$$

  until $\theta_{s+1} \approx \theta_s$.

- If there is more than one parameter, use the multivariate version of Newton-Raphson.

Example.

Compute the ML for $X \sim \text{Bernoulli}(p)$.

In this case we use $p_X(X_i) = p^{X_i}(1-p)^{1-X_i}$, so the log likelihood is

$$\ln L = \sum_{i=1}^{n} \ln\left(p^{X_i}(1-p)^{1-X_i}\right) = \sum_{i=1}^{n} \left(X_i \ln(p) + (1 - X_i)\ln(1-p)\right)$$

which exists when $0 < p < 1$.

Now differentiate:

$$\frac{\partial \ln L}{\partial p} = \sum_{i=1}^{n} \frac{X_i}{p} - \frac{1 - X_i}{1-p}$$

so $\hat{p}$ satisfies the score equation

$$\sum_{i=1}^{n} \frac{X_i}{\hat{p}} - \frac{1 - X_i}{1-\hat{p}} = 0$$

Now solving:

$$\sum_{i=1}^{n} \frac{X_i}{\hat{p}} - \frac{1 - X_i}{1-\hat{p}} = 0$$

$$(1-\hat{p})\sum_{i=1}^{n} X_i = \hat{p}\sum_{i=1}^{n}(1 - X_i)$$

$$\sum_{i=1}^{n} X_i - \hat{p}\sum_{i=1}^{n} X_i = n\hat{p} - \hat{p}\sum_{i=1}^{n} X_i$$

$$\sum_{i=1}^{n} X_i = n\hat{p}$$

thus

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$$

2.4.2 ML comments

To obtain the ML, we **must** know $f_X(\cdot)$ (e.g., a normal distribution) and the $X_1, \ldots, X_n$ **must** be i.i.d. with probability density function $f_X(\cdot)$. Contrast this with OLS, where we need only know some moments, and do not require that the sample be identically distributed.

However, the additional assumptions needed for ML make it a very powerful estimator. ML is used to benchmark other estimators based on weaker assumptions. For some models it is indispensable.

Example.

For example, regression models with binary outcome data (Probit and Logit models).

2.4.3 ML and regression

ML can be modified to allow for a regression setting with an outcome Y_i and a regressor X_i.

To do this, we must specify $f_{Y|X}(Y_i \mid X_i)$ instead of $f_X(X_i)$, so the likelihood is

$$L(\theta \mid (Y_1, X_1), \ldots, (Y_n, X_n)) = \prod_{i=1}^{n} f_{Y|X}(Y_i \mid X_i)$$

Example.

For example, we can use

$$Y_i \mid X_i \sim \text{Normal}(\beta_1 + \beta_2 X_i, \sigma^2)$$

which corresponds to the simple linear regression model where we additionally assume U_i are i.i.d. $\text{Normal}(0, \sigma^2)$ for $i = 1, \ldots, n$.

In this case, the ML and OLS estimators of β_1, β_2 are identical!

ML is much more powerful than just linear regression!

Example.

For example, we can model count outcomes using

$$Y_i \mid X_i \sim \text{Poisson}(\lambda(X_i))$$

- Since the Poisson parameter is $\lambda > 0$, a common parameterisation is

$$\lambda(X_i) = \exp(\gamma_1 + \gamma_2 X_i)$$

- Y_i could be annual nights spent in hospital and X_i could be health insurance status.

Example.

In the same way, one can model binary outcomes (e.g., employed/unemployed) by using

$$Y_i \mid X_i \sim \text{Bernoulli}(p(X_i))$$

Theorem 2.4.3: ML properties

ML is

- Consistent: $\hat{\theta} \xrightarrow{p} \theta$
- Asymptotically unbiased: $\lim_{n\to\infty} E(\hat{\theta}) = \theta$
- Asymptotically normally distributed: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \text{Normal}(0, nI(\theta)^{-1})$
- Asymptotically normally distributed:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \text{Normal}(0, nI(\theta)^{-1})$$

  so that

$$\hat{\theta} \overset{a}{\sim} \text{Normal}(\theta, I(\theta)^{-1})$$

- Asymptotically efficient: because its asymptotic variance is $I(\theta)^{-1} = \text{CRLB}$

However, ML can sometimes be biased.

2.5 Cramer-Rao Lower Bound

The CRLB provides a lower bound on the variance of any unbiased estimator.

Theorem 2.5.1

We can thus determine the efficiency of an unbiased estimator by comparing its variance to the CRLB. It allows us to say that an unbiased estimator is the 'best' if its variance is equal to the CRLB.

Recall

$$E((\hat{\theta} - \theta)^2) = (E(\hat{\theta}) - \theta)^2 + V(\hat{\theta})$$

Since the MSE of an unbiased estimator equals its variance, the CRLB can be thought of as a lower bound on the MSE of an unbiased estimator.

Theorem 2.5.2

If θ comprises one parameter, under regularity conditions we have the following results:

- Expectation of the score:

$$\mathbb{E}\left(\frac{\partial \ln L}{\partial \theta}\right) = 0$$

- CRLB: For an unbiased estimator $\hat{\theta}$,

$$\mathbb{V}(\hat{\theta}) \geq \frac{1}{\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)}$$

Example.

> Does the ML estimator of $X \sim \text{Bernoulli}(p)$ attain the CRLB?

To compute the CRLB we need to compute $\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)$. Recall

$$\frac{\partial \ln L}{\partial p} = \sum_{i=1}^{n} \left(\frac{X_i}{p} - \frac{1 - X_i}{1 - p}\right)$$

so

$$\frac{\partial^2 \ln L}{\partial p^2} = \sum_{i=1}^{n} \left(-\frac{X_i}{p^2} - (-1) \left(\frac{1 - X_i}{(1-p)^2} \right) \right) = -\sum_{i=1}^{n} \frac{X_i}{p^2} + \frac{1 - X_i}{(1-p)^2}$$

thus

$$\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial p^2} \right) = -\sum_{i=1}^{n} \left(\frac{p}{p^2} + \frac{1-p}{(1-p)^2} \right) = -\sum_{i=1}^{n} \left(\frac{1}{p} + \frac{1}{1-p} \right) = -\sum_{i=1}^{n} \frac{1}{p(1-p)} = -\frac{n}{p(1-p)}$$

So the CRLB is

$$-\frac{1}{\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial \theta^2} \right)} = \frac{p(1-p)}{n}$$

Recall that the ML was $\hat{p} = \bar{X}$.

We have previously shown that if X has expectation μ and variance σ^2, then

$$\mathbb{E}(\bar{X}) = \mu, \quad \mathbb{V}(\bar{X}) = \frac{\sigma^2}{n}$$

Since $X \sim \text{Bernoulli}(p)$, in this case $\mu = p$ and $\sigma^2 = p(1-p)$, thus

$$\mathbb{E}(\hat{p}) = p, \quad \mathbb{V}(\hat{p}) = \frac{p(1-p)}{n}$$

Thus, the ML estimator is unbiased and attains the CRLB. Hence it is the best (most efficient) unbiased estimator.

Theorem 2.5.3

More generally, if $\theta = (\theta_1, \theta_2, \ldots, \theta_q)$,

- Expectation of the score:
$$\mathbb{E}\left(\frac{\partial \ln L}{\partial \theta_j} \right) = 0, \quad j = 1, \ldots, q$$
- CRLB: For an unbiased estimator $\hat{\theta}$,
$$\mathbb{V}(\hat{\theta}) \succeq -\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial \theta \partial \theta'} \right)^{-1}$$
- Note that $\mathbb{V}(\hat{\theta})$ and $\frac{\partial^2 \ln L}{\partial \theta \partial \theta'}$ are $q \times q$ matrices.
- The matrix
$$-\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial \theta \partial \theta'} \right) = I(\theta)$$
is the *information matrix*, so $\mathbb{V}(\hat{\theta}) \succeq I(\theta)^{-1}$.

When $q = 1$,

$$\frac{\partial^2 \ln L}{\partial \theta \partial \theta'} = \frac{\partial^2 \ln L}{\partial \theta^2}$$

so

$$I(\theta) = -\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial \theta^2} \right)$$

When $q = 2$,

$$\mathbb{V}(\hat{\theta}) = \begin{pmatrix} \mathbb{V}(\hat{\theta}_1) & \text{COV}(\hat{\theta}_1, \hat{\theta}_2) \\ \text{COV}(\hat{\theta}_1, \hat{\theta}_2) & \mathbb{V}(\hat{\theta}_2) \end{pmatrix}$$

and

$$\frac{\partial^2 \ln L}{\partial\theta\partial\theta'} = \begin{pmatrix} \frac{\partial^2 \ln L}{\partial\theta_1^2} & \frac{\partial^2 \ln L}{\partial\theta_1\partial\theta_2} \\ \frac{\partial^2 \ln L}{\partial\theta_1\partial\theta_2} & \frac{\partial^2 \ln L}{\partial\theta_2^2} \end{pmatrix}$$

thus

$$I(\theta) = \begin{pmatrix} -\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial\theta_1^2}\right) & -\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial\theta_1\partial\theta_2}\right) \\ -\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial\theta_1\partial\theta_2}\right) & -\mathbb{E}\left(\frac{\partial^2 \ln L}{\partial\theta_2^2}\right) \end{pmatrix}$$

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then (if it exists)

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

2.6 Confidence intervals

When we estimate a parameter, we make a mistake. That is $\hat{\theta} \neq \theta$. In fact, because $\hat{\theta}$ usually has a continuous distribution, $\text{Prob}(\hat{\theta} = \theta) = 0$. So it doesn't make sense to report $\hat{\theta}$ as if it were the truth. Instead, we need to summarise how much uncertainty there is regarding the true value of θ. We do this by constructing a confidence interval.

2.6.1 Confidence intervals

Definition 2.6.1

Let θ be a parameter. Given a random sample, let $T_1 \leq T_2$ be two statistics (i.e. functions of the sample) which satisfy either

$$\text{Prob}(T_1 \leq \theta \leq T_2) = 1 - \alpha \quad \text{or} \quad \lim_{n\to\infty} \text{Prob}(T_1 \leq \theta \leq T_2) = 1 - \alpha$$

where α is a constant which does not depend on θ. Then the random interval (T_1, T_2) is a $1 - \alpha$ *confidence interval* for θ.

Given data, just like we observe an estimate (a realisation of $\hat{\theta}$), we can observe an estimate (t_1, t_2) of (T_1, T_2). The interpretation is that, if we could (hypothetically) observe many datasets of size n, θ would lie between t_1 and t_2 in $100 \times (1 - \alpha)\%$ of the datasets.

It is straightforward to modify the definition above for any function $g(\theta)$ (just replace θ with $g(\theta)$).

2.6.2 Pivotal statistics

Definition 2.6.2

Let θ be a parameter. Given a random sample, let $Q(\theta)$ be a random variable the distribution of which does not depend on θ. Then $Q(\theta)$ is a *pivotal statistic*.

We (usually) construct confidence intervals using a pivotal statistic. Since we know the distribution of $Q(\theta)$, we can find constants $q_1 \leq q_2$ such that either

$$\text{Prob}(q_1 \leq Q(\theta) \leq q_2) = 1 - \alpha \quad \text{or} \quad \lim_{n\to\infty} \text{Prob}(q_1 \leq Q(\theta) \leq q_2) = 1 - \alpha$$

Then we can rearrange the inequalities to isolate θ (or $g(\theta)$).

$$\text{Prob}(q_1 \leq Q(\theta) \leq q_2) = 1 - \alpha$$

We can use the known distribution of $Q(\theta)$ to choose any q_1, q_2 satisfying the above. Usually we use an

equi-tailed approach in which we find q_1, q_2 solving

$$\text{Prob}(Q(\theta) < q_1) = \text{Prob}(Q(\theta) > q_2) = \frac{\alpha}{2}$$

Or we may choose q_1, q_2 to minimize $q_2 - q_1$, which results in the narrowest possible confidence interval. $Q(\theta)$ usually follows a Normal$(0, 1)$ distribution, in which case these approaches coincide.

Small sample

To find a pivotal statistic, we usually need to know the distribution of some estimator $\hat{\theta}$. We rarely know the small sample distribution of $\hat{\theta}$.

A leading example of a case in which we do is when we know that the random sample is normal. Recall that in this case we know that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \text{Normal}(0, 1), \quad \frac{\overline{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}, \quad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

Example.

> Suppose that we want a 0.95 confidence interval for σ^2.

We can use the pivotal statistic

$$Q(\sigma^2) = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

For the equi-tailed confidence interval, q_1 is the 0.025 percentile of χ^2_{n-1} and q_2 is the 0.975 percentile of χ^2_{n-1} and

$$\text{Prob}\left(q_1 \leq \frac{(n-1)s^2}{\sigma^2} \leq q_2\right) = 0.95$$

Divide by $(n-1)s^2 > 0$:

$$\text{Prob}\left(\frac{q_1}{(n-1)s^2} \leq \frac{1}{\sigma^2} \leq \frac{q_2}{(n-1)s^2}\right) = 0.95$$

Now take the reciprocal:

$$\text{Prob}\left(\frac{(n-1)s^2}{q_1} \geq \sigma^2 \geq \frac{(n-1)s^2}{q_2}\right) = 0.95$$

So the 0.95 confidence interval is:

$$\left(\frac{(n-1)s^2}{q_2}, \frac{(n-1)s^2}{q_1}\right)$$

- *This does not take the familiar form $\hat{\theta} \pm q\sqrt{V(\hat{\theta})}$*
- *It is a multiplicative rather than additive function of the estimator s^2*
- *Notice that the lower bound will always be positive*

Large sample

To find a pivotal statistic, we usually need to know the distribution of some estimator $\hat{\theta}$. We rarely know the small sample distribution of $\hat{\theta}$.

So we use the large sample distribution of $\hat{\theta}$ as an approximation. Many estimators (e.g., ML, OLS) have an asymptotic normal distribution, so pivotal statistics are typically based on the normal distribution.

Example.

> Given a random sample from a distribution with mean μ and variance σ^2, find a 0.95 confidence interval for μ.

Using CLT, the pivotal statistic satisfies

$$\lim_{n\to\infty} \text{Prob}\left(-1.96 \leq \frac{\overline{X}-\mu}{s/\sqrt{n}} \leq 1.96\right) = 0.95$$

So a 0.95 confidence interval is

$$\left(\overline{X} - 1.96\frac{s}{\sqrt{n}},\ \overline{X} + 1.96\frac{s}{\sqrt{n}}\right)$$

Since it is based on CLT, we can use this approach for a random sample from ***any*** *distribution*

Recall

$$\lim_{n\to\infty} \text{Prob}\left(\overline{X} - 1.96\frac{s}{\sqrt{n}} \leq \mu \leq \overline{X} + 1.96\frac{s}{\sqrt{n}}\right) = 0.95$$

For large n we know that our confidence interval is reliable. For small n it can be unreliable and is usually too narrow. A common correction is to use the t_{n-1} distribution instead of Normal$(0,1)$. Since the t_{n-1} has fatter tails, it makes the interval wider. There is no formal justification for this, but it works well in practice if the random sample is "close" to normal. The correction is widely used, for example for ML and OLS. As $n \to \infty$, $t_{n-1} \to$ Normal$(0,1)$ so it still works well when n is large.

2.7 Hypothesis testing

Hypothesis testing allows us to learn about economic primitives.

Example.

> For example, we might estimate the parameters of the Cobb-Douglas production function $Y = B_1 L^{\beta_2} U$ by estimating $y = \beta_1 + \beta_2 l + u$ using OLS, where lower case letters are expressed in natural logarithms. Then we could test whether there are constant returns to scale (i.e. $\beta_2 = 1$).

There are two types of tests:

- Distributional tests (e.g. did my data come from a normal distribution)
- Parametric tests on parameters (e.g. $\theta = 0$)

We begin with a 'null' or 'maintained' hypothesis H_0. We also specify an 'alternative' hypothesis H_1 which is what we revert to if H_0 is rejected. $H_1 = H_0^c$ is common. We will maintain H_0 unless there is overwhelming evidence to the contrary (e.g. H_0: innocent and H_1: guilty).

Example.

> Given a random sample, test $\mu = \mu_0$.

First specify

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \neq \mu_0$$

We start by believing that H_0 is true. If H_0 is true then we know that

$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}} \overset{a}{\sim} \text{Normal}(0,1)$$

Now we look at our data, construct t (the realisation of T in our data) and ask ourselves whether t is 'reasonable' given H_0 is true. A value 'close' to zero would be 'reasonable', but a value 'far' from zero would not. So if we observe t close to zero then we have no reason to question H_0. Conversely, if we observe t far from zero then we might question whether H_0 is really true.

Now we have to decide what 'reasonable' means. That is, we have to decide on a cut-off c which defines 'close' to zero. Logic then dictates that we reject H_0 if $|t| > c$ and do not reject H_0 otherwise.

More formally, We need to define a *rejection region* R such that we reject H_0 if $(x_1, \ldots, x_n) \in R$. Instead of working with $R \subseteq \mathbb{R}^n$, we usually prefer to construct a test-statistic $T(X_1, \ldots, X_n)$ and a region $R \subseteq \mathbb{R}$ such that we reject H_0 if $t(x_1, \ldots, x_n) \in R$.

In this example,

$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}, \qquad R = (-\infty, -c) \cup (c, \infty)$$

Definition 2.7.1

A hypothesis is *simple* if it specifies values of all of the parameters. Otherwise it is *composite*.

For the rest of the lecture we focus on simple null and alternative hypotheses.

Example.

For example, $H : \mu = 0$ is simple whereas $H : \mu > 0$ is composite

2.7.1 Two types of error

Definition 2.7.2

When conducting a test, we can make two mistakes:

- *Type I error:* Reject a true null
- *Type II error:* Fail to reject a false null

The likelihood of making such errors will be determined by the rejection region.
Formally, we use

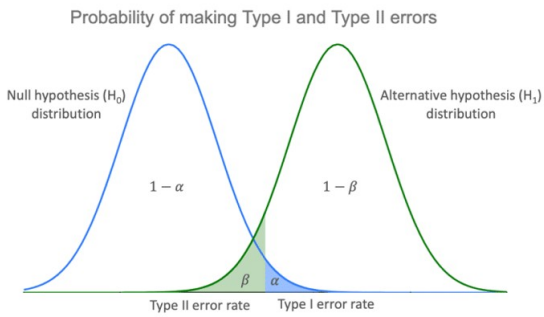
$$\alpha = \text{Prob}(T \in R \mid H_0), \qquad \beta = \text{Prob}(T \in R^c \mid H_1)$$

and we would like to design the test so that they are small.

Definition 2.7.3

The power of a test is

$$\text{Prob}(T \in R \mid H_1) = 1 - \beta$$



Suppose that the rejection region is $R = (c, \infty)$ and we have the distributions $T \mid H_0$ and $T \mid H_1$. Notice that we cannot decrease α without increasing β and vice versa.

Example.

Consider a random sample of size $n = 1$ from

$$f_X(x) = \begin{cases} 1 - \theta + x & \theta - 1 \leq x \leq \theta \\ 1 + \theta - x & \theta \leq x \leq \theta + 1 \end{cases}$$

and $H_0 : \theta = 0$, $H_1 : \theta = 1$. Suppose that $T = X_1$ and the rejection region is $R = (c, \infty)$ for $c \in [0, 1]$. Compute the type I and II error probabilities.

For the type I error we need

$$f_{X|H_0}(x|H_0) = \begin{cases} 1+x & -1 \le x \le 0 \\ 1-x & 0 \le x \le 1 \end{cases}$$

so

$$\alpha = \text{Prob}(T > c|H_0) = \int_c^1 (1-t)dt = \frac{1}{2}(1-c)^2$$

For the type II error we need

$$f_{X|H_1}(x|H_1) = \begin{cases} x & 0 \le x \le 1 \\ 2-x & 1 \le x \le 2 \end{cases}$$

so

$$\beta = \text{Prob}(T \le c|H_1) = \int_0^c t dt = \frac{1}{2}c^2$$

Hence

$$\alpha = \frac{1}{2}(1-c)^2, \quad \beta = \frac{1}{2}c^2$$

We can also find a relationship between α and β

$$\alpha = \frac{1}{2}\left(1 - \sqrt{2\beta}\right)^2$$

which is decreasing in β because $\beta = \frac{1}{2}c^2 \in [0, 1/2]$.

Definition 2.7.4

A test is *admissible* if there is no other test with either α smaller and β no larger or β smaller and α no larger.

The set of admissible tests plotted on the (α, β) plane is a continuous, monotonically decreasing, convex function which starts at a point in $[0, 1]$ on the β axis and ends at a point in $[0, 1]$ on the α axis.

2.7.2 Choosing a test

Given a test-statistic T, how should we choose a rejection region R? There is a trade-off between α and β. Holding α constant we would like to choose the rejection region with smallest β, or vice versa.

By convention we hold α constant and usually take $\alpha \le 0.1$. This means that we are quite attached to H_0 (innocence) and only reject it given overwhelming evidence to the contrary ('beyond reasonable doubt'). We'll return to the issue of how to choose the test itself later.

Example.

> Recall $H_0 : \mu = \mu_0$ vs $H_1 : \mu \ne \mu_0$ and
>
> $$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}} \overset{a}{\sim} \text{Normal}(0, 1), \qquad R = (-\infty, -c) \cup (c, \infty)$$

Now fix $\alpha \in (0, 1)$ and find $c > 0$ which solves $\text{Prob}(T \in R|H_0) = \alpha$, or equivalently

$$\text{Prob}(T \in R^c|H_0) = 1 - \alpha \Rightarrow \text{Prob}(-c \le Z \le c) = 1 - \alpha$$

where $Z \sim \text{Normal}(0, 1)$. So c should be the $1 - \alpha/2$ percentile of $\text{Normal}(0, 1)$. If $\alpha = 0.05$ then $c = 1.96$.

Definition 2.7.5

Recall that if $\mu_0 = 0$ this is a test of statistical significance. If $t \in R$, we say that μ is *statistically significant*. Otherwise μ is said to be *statistically insignificant*.

Definition 2.7.6

Because it is based on the standard normal distribution, this test is commonly referred to as a *z-test*

We used an asymptotic approximation to choose c, so $\text{Prob}(T \in R|H_0) \neq \alpha$ but $\lim_{n\to\infty} \text{Prob}(T \in R|H_0) = \alpha$

Definition 2.7.7

If n is small it is preferable to use t_{n-1} instead: We call this a *t-test*

The argument is the same as for confidence intervals. It yields $\text{Prob}(T \in R|H_0)$ closer to α for small n and still delivers $\lim_{n\to\infty} \text{Prob}(T \in R|H_0) = \alpha$

Lemma 2.7.8: Neyman-Pearson Lemma

Given a random sample $X_1, \ldots, X_n$, in testing $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$, the most powerful size α test satisfies

$$R = \left\{ x_1, \ldots, x_n : \frac{L(\theta_0 \mid x_1, \ldots, x_n)}{L(\theta_1 \mid x_1, \ldots, x_n)} < c \right\}$$

where c is determined by $\text{Prob}[(X_1, \ldots, X_n) \in R \mid H_0] = \alpha$ provided that such c exists.

- *A test-statistic and rejection region can be derived from the inequality*
- *It is often useful to take logs of both sides of the inequality in R*
- *It is related to the likelihood ratio test (more on this later)*

Example.

> Given a random sample of size 1 from a *Bernoulli*(p) distribution, derive R for the test $H_0 : p = p_0$ vs $H_1 : p = p_1$

We have one observation, denoted x. Applying the NP Lemma:

$$\frac{L(\theta_0 \mid x)}{L(\theta_1 \mid x)} < c$$

$$\frac{p_0^x (1-p_0)^{1-x}}{p_1^x (1-p_1)^{1-x}} < c$$

$$x \left(\ln \left[\frac{p_0}{1-p_0} \right] - \ln \left[\frac{p_1}{1-p_1} \right] \right) + \ln \left[\frac{1-p_0}{1-p_1} \right] < \ln c$$

$$ax + b < d$$

If $p_0 > p_1$ then $a > 0$ so $x < (d-b)/a$. If $p_0 < p_1$ then $a < 0$ so $x > (d-b)/a$. Now let's proceed for the case $p_0 > p_1$. We need to choose c (hence d because $d = \ln c$) so that

$$\text{Prob}(X \in R \mid H_0) = \alpha \qquad \text{or} \qquad \text{Prob}\left(X < \frac{d-b}{a} \mid H_0 \right) = \alpha$$

Under H_0, $X \sim \text{Bernoulli}(p_0)$, so:

$$\Pr(X < e \mid H_0) = \begin{cases} 0, & e \leq 0, \\ 1 - p_0, & 0 < e \leq 1, \\ 1, & e > 1. \end{cases}$$

This means that we can only allow for three values of α, two of which are trivial! For other values of α there does not exist a c for the NP Lemma. Let's now take $\alpha = 1 - p_0$ so

$$\frac{d-b}{a} = 1 \Rightarrow d = a + b \Rightarrow c = e^{a+b}$$

Definition 2.7.9

The *power function* is $Q(\theta) = \text{Prob}(T \in R \mid \theta)$.
So for $H_0 : \theta = \theta_0$ vs $H_0 : \theta = \theta_1$ we have

$$\alpha = Q(\theta_0), \qquad 1 - \beta = Q(\theta_1)$$

Definition 2.7.10

Let $Q_1(\cdot)$ and $Q_2(\cdot)$ be the power functions of two tests of $H_0 : \theta = \theta_0$ vs $H_1 : \theta \in \Theta_1$. Then the first test is *uniformly more powerful* than the second if

$$Q_1(\theta_0) = Q_2(\theta_0) = \alpha, \qquad Q_1(\theta_1) \geq Q_2(\theta_1), \quad \forall \theta_1 \in \Theta_1$$

- *It is uniformly more powerful because it is more powerful no matter the value of $\theta_1 \in \Theta_1$*
- *A test is uniformly most powerful if it is uniformly more powerful than any other test*
- *A uniformly most powerful test may not exist, and if it does, it may not be unique*

2.7.3 Likelihood ratio test

If a uniformly most powerful test exists, it is the LR test. If a uniformly most powerful test does not exist, the LR test has favourable asymptotic properties. The LR test is a generalisation of the Neyman-Pearson Lemma.

Definition 2.7.11: LR test

The LR test for $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$ has rejection region

$$R = \left\{ x_1, \ldots, x_n : \frac{\sup_{\theta \in \Theta_0} L(\theta \mid x_1, \ldots, x_n)}{\sup_{\theta \in \Theta_0 \cup \Theta_1} L(\theta \mid x_1, \ldots, x_n)} < c \right\}$$

where c is determined by $\text{Prob}[(X_1, \ldots, X_n) \in R \mid H_0] = \alpha$ provided that such c exists.

- *We can test a composite null versus a composite alternative*
- *We may not always be able to find such c*

Example.

> Given a random sample from a Normal(μ, σ^2) distribution and known σ^2, conduct a LR test of $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$.

The rejection region is

$$\frac{\sup_{\mu=\mu_0} L(\mu \mid x_1, \ldots, x_n)}{\sup_{\mu} L(\mu \mid x_1, \ldots, x_n)} = \frac{L(\mu_0 \mid x_1, \ldots, x_n)}{\sup_{\mu} L(\mu \mid x_1, \ldots, x_n)} < c$$

The likelihood is

$$L(\mu \mid x_1, \ldots, x_n) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \right)$$

So,

$$\sup_{\mu} L(\mu \mid x_1, \ldots, x_n) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \inf_{\mu} \sum_{i=1}^n (x_i - \mu)^2\right)$$

so we need to find the μ which minimizes $\sum_{i=1}^n (x_i - \mu)^2$, which is $\overline{x}$. The rejection region is

$$\frac{L(\mu_0 \mid x_1, \ldots, x_n)}{L(\overline{x} \mid x_1, \ldots, x_n)} < c$$

$$\frac{\exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu_0)^2\right)}{\exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \overline{x})^2\right)} < c$$

$$\sum_{i=1}^n (x_i - \overline{x})^2 - \sum_{i=1}^n (x_i - \mu_0)^2 < 2\sigma^2 \ln c$$

$$\left(\sum_{i=1}^n x_i^2 - n\overline{x}^2\right) - \left(\sum_{i=1}^n x_i^2 - 2n\mu_0\overline{x} + n\mu_0^2\right) < 2\sigma^2 \ln c$$

$$-n\overline{x}^2 + 2n\mu_0\overline{x} - n\mu_0^2 < 2\sigma^2 \ln c \Rightarrow -n(\overline{x} - \mu_0)^2 < 2\sigma^2 \ln c \Rightarrow \left(\frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}}\right)^2 > -2\ln c$$

The rejection region is

$$\left(\frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}}\right)^2 > -2\ln c = d^2 \quad \text{or equivalently} \quad \left|\frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}}\right| > d$$

For $d > 0$, now we need to solve $\text{Prob}[(X_1, \ldots, X_n) \in R \mid H_0] = \alpha$.
Under H_0,

$$\frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim \text{Normal}(0, 1)$$

Hence d is the $1 - \alpha/2$ percentile of $\text{Normal}(0, 1)$.

Theorem 2.7.12

If we cannot find a suitable c, we can use the result

$$-2\ln\Lambda \xrightarrow{d} \chi_d^2$$

where d is the degrees of freedom and

$$\Lambda = \frac{\sup_{\theta \in \Theta_0} L(\theta \mid X_1, \ldots, X_n)}{\sup_{\theta \in \Theta_0 \cup \Theta_1} L(\theta \mid X_1, \ldots, X_n)}$$

We have $T = -2\ln\Lambda$ and $R = (c, \infty)$ where c is the $1 - \alpha$ percentile of χ_d^2. Then we have

$$\lim_{n \to \infty} \text{Prob}(T \in R \mid H_0) = \alpha$$

The asymptotic LR test can be unreliable if n is small.

Example.

A leading case is a test of the joint hypothesis

$$H_0 : \theta_1 = \theta_{1,0}, \theta_2 = \theta_{2,0}, \ldots, \theta_d = \theta_{d,0} \quad \text{vs} \quad H_1 : H_0^c$$

in which case

$$\Lambda = \frac{L(\theta_0 \mid X_1, \ldots, X_n)}{L(\hat{\theta} \mid X_1, \ldots, X_n)}$$

where $\hat{\theta}$ is the ML estimator and

$$-2\ln\Lambda \xrightarrow{d} \chi^2_d$$

If $\theta_{1,0} = \theta_{2,0} = \cdots = 0$ this is a test of *joint statistical significance.*

- If $t \in R$ we say that $\theta_1, \theta_2, \ldots, \theta_d$ are jointly statistically significant
- Otherwise $\theta_1, \theta_2, \ldots, \theta_d$ are jointly statistically insignificant

2.7.4 Wald test

Recall that to construct the likelihood function we need a random sample from a known distribution. So to conduct the LR test we need a random sample from a known distribution. The Wald test does not require knowledge of the distribution, so is more generally applicable in practice.

Example.

Recall $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$ and

$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}} \overset{a}{\sim} \text{Normal}(0,1), \qquad R = (-\infty, -c) \cup (c, \infty)$$

$$\lim_{n\to\infty} \text{Prob}(T \in R \mid H_0) = \alpha$$

where c is the $1 - \alpha/2$ percentile of Normal$(0,1)$.

Consider now the identical test

$$\tilde{T} = T^2 \overset{a}{\sim} \chi^2_1, \qquad \tilde{R} = (c^2, \infty)$$

This is a *Wald test.* Note also that c^2 is the $1-\alpha$ percentile of χ^2_1. If $\alpha = 0.05$ then $c = 1.96$ and $c^2 = 1.96^2 = 3.84$. So this z-test is a special case of a Wald test.

Recall that if n is small it is preferable to use t_{n-1} instead of Normal$(0,1)$ for the distribution of T. If $T \overset{a}{\sim} t_{n-1}$ then $\tilde{T} = T^2 \overset{a}{\sim} F_{1,n-1}$. So if n is small it is preferable to use $F_{1,n-1}$ instead of χ^2_1. It yields $\text{Prob}(\tilde{T} \in \tilde{R} \mid H_0)$ closer to α for small n and still delivers $\lim_{n\to\infty} \text{Prob}(\tilde{T} \in \tilde{R} \mid H_0) = \alpha$ because as $n \to \infty$, $F_{1,n-1} \to \chi^2_1$. We call this an *F-test.*

Definition 2.7.13

The Wald test-statistic for $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ was

$$\tilde{T} = \left(\frac{\overline{X} - \mu_0}{s/\sqrt{n}}\right)^2 = (\overline{X} - \mu_0)\left(\frac{s^2}{n}\right)^{-1}(\overline{X} - \mu_0) \overset{a}{\sim} \chi^2_1$$

More generally, if $\theta = (\theta_1, \theta_2, \ldots, \theta_q)$ to test

$$H_0 : h_1(\theta) = 0,\ h_2(\theta) = 0, \ldots, h_d(\theta) = 0 \quad \text{vs} \quad H_1 : H_0^c$$

we can use the *quadratic form*

$$\tilde{T} = \mathbf{h}(\hat{\theta})'\hat{V}[\mathbf{h}(\hat{\theta})]^{-1}\mathbf{h}(\hat{\theta}) \overset{a}{\sim} \chi^2_d, \qquad \mathbf{h}(\hat{\theta}) = \begin{pmatrix} h_1(\hat{\theta}) \\ h_2(\hat{\theta}) \\ \vdots \\ h_d(\hat{\theta}) \end{pmatrix}$$

The rejection region is $\tilde{R} = (c^2, \infty)$ where c^2 is the $1-\alpha$ percentile of χ^2_d.

For small n it is preferable to use $dF_{d,n-q}$ instead of χ^2_d. As $n \to \infty$, $dF_{d,n-q} \to \chi^2_d$.

Example.

> Suppose that we have an ML estimate of θ_1, θ_2 given by
> $\hat{\theta}_1 = \hat{\theta}_2 = 1/2$ and $\widehat{V}(\hat{\theta}_1) = 1$, $\widehat{V}(\hat{\theta}_2) = 4$ and $\widehat{\mathrm{COV}}(\hat{\theta}_1, \hat{\theta}_2) = 0$.
> Test $H_0 : \theta_1 = 1, \theta_2 = 0$ vs $H_1 : H_0^c$.

We have

$$\mathbf{h}(\hat{\theta}) = \begin{pmatrix} \hat{\theta}_1 - 1 \\ \hat{\theta}_2 \end{pmatrix}, \qquad \widehat{V}[\mathbf{h}(\hat{\theta})] = \widehat{V}\left[\begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{pmatrix}\right] = \begin{pmatrix} \widehat{V}(\hat{\theta}_1) & \widehat{\mathrm{COV}}(\hat{\theta}_1, \hat{\theta}_2) \\ \widehat{\mathrm{COV}}(\hat{\theta}_1, \hat{\theta}_2) & \widehat{V}(\hat{\theta}_2) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

So

$$\begin{aligned} \tilde{t} &= \begin{pmatrix} 1/2 - 1 \\ 1/2 \end{pmatrix}' \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}^{-1} \begin{pmatrix} 1/2 - 1 \\ 1/2 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix}' \begin{pmatrix} 1 & 0 \\ 0 & 1/4 \end{pmatrix} \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix} \\ &= (-1/2)^2 \cdot 1 + 2 \cdot 0 \cdot (-1/2) \cdot (1/2) + (1/2)^2 \cdot (1/4) = 0.3125 \end{aligned}$$

There are two restrictions, so the rejection region for $\alpha = 0.05$ is $\tilde{R} = (5.99, \infty)$ because 5.99 is the $1 - \alpha$ percentile of χ^2_2.

Since $\tilde{t} = 0.3125$ does not lie in the rejection region, we fail to reject H_0 for $\alpha = 0.05$.